Master Research Execution Plan

English offline productivity commands on a resource constrained Android phone

Denys Menfredy | 20 October to 14 December 2026 | Version 1.1

Purpose and completion target

Complete a reproducible empirical study on the Realme 7 Pro with 8 GB RAM, supported by a working deployment, controlled experiments, a thesis chapter draft, and a submission-ready manuscript. The eight-week target covers research and writing readiness. Degree completion also depends on the university calendar, supervisor review, examination, and administrative requirements.

Working thesis title

Design and Evaluation of a Lightweight On-Device Language Model with Domain-Specific Adaptation

This study adapts an existing pretrained model. It does not train a new foundation model or claim a new architecture. If the university requires a narrower title, use Domain Adaptation and Quantization of a Small Language Model for Offline Mobile Productivity Commands.

Decisions already established

- Primary hardware: Realme 7 Pro, 8 GB physical RAM, Snapdragon 720G and Adreno 618. Record the exact Android version and firmware build in week one [1].
- Required execution path: CPU inference. GPU acceleration is an optional feasibility experiment. NPU execution and additional devices are outside the minimum study.
- Primary model remains open: compare Qwen3-0.6B with Qwen3.5-0.8B, then select one before full training [2, 3].
- Planning capacity: 20 hours per week for eight weeks, totaling 160 researcher hours, plus unattended training and inference time.

Application scope and first supervisor review

The primary language is English. The application converts typed calendar, alarm, and list requests into structured proposed actions without network access. Audit the English MASSIVE subset before freezing its coverage. PT-BR is an optional extension after the core study; European Portuguese data must not be relabeled as Brazilian Portuguese. Agree on the contribution, university requirements, compute access, and publication route with the supervisor.

2 Research questions and success criteria

Primary question

For a fixed small language model on the Realme 7 Pro, how does domain-specific LoRA adaptation affect joint intent and slot correctness for English offline productivity commands after 4-bit and 8-bit quantization, and which configurations meet predefined latency and memory requirements?

Question	Comparison and evidence
RQ1 Adaptation	Untuned versus adapted at matched deployment precision; joint semantic exact match differences with uncertainty.
RQ2 Quantization	High-precision workstation reference versus Q8_0 and Q4_K_M; quantify quality loss and whether adaptation gains persist.
RQ3 Deployment	Quality, p95 latency, peak process memory, failures, and sustained performance on the same phone.
RQ4 Optional policy	Heuristic selection versus fixed Q4 and fixed Q8 under controlled memory pressure; include switching costs.

Separate research success from model success

Research success means the protocol is executed, uncertainty and failures are reported, and the conclusions follow from evidence. The model need not improve. A null or negative result is acceptable when the baseline is credible and the experiment is sufficiently sensitive.

Deployment success means a configuration satisfies the application requirements. Define those requirements with the supervisor before inspecting final test results. Do not relax requirements afterward to make a configuration appear successful.

Requirements to freeze by the end of week two

- Minimum acceptable quality: [joint semantic exact match threshold]. Smallest practically meaningful adaptation gain: [percentage points].
- Maximum acceptable p95 request latency: [seconds]. Memory budget: [MiB peak process memory using the chosen measurement].
- Sustained workload: [duration, request pattern, timeout]. Acceptable failure rate and performance degradation: [limits].

If no product requirement exists, report a quality-latency-memory frontier instead of inventing a universal deployment threshold. Any example numerical threshold is a planning choice, not an expected result.

Claim boundaries

The primary task evaluates semantic parsing, not speech recognition, factual QA, or successful real-world action execution. A single-phone study supports a case study for that software stack. Privacy is motivated by local processing, not established by a formal security evaluation. Public benchmark contamination and limited domain coverage must be discussed.

3 Scope and model selection

Mandatory study and optional extensions

Mandatory	Optional after core completion
One domain and one primary model	A second model trained end to end
LoRA with three final training seeds	Training-data scaling or an independent robustness set
CPU Q4 and Q8 comparisons	Verified GPU offload comparison
Rule-based and joint intent and slot encoder baselines	Heuristic configuration selection
Repeated device measurements and manuscript	Energy measurements or PT-BR after the core study

Week one model pilot

Evaluate Qwen3-0.6B and Qwen3.5-0.8B with text-only inputs, explicit reasoning disabled where configurable, and recorded templates. Qwen3.5-0.8B uses a hybrid architecture and includes a vision encoder; validate its text-only training and export path rather than assuming compatibility from parameter count [2, 3].

- Use the same stratified validation panel of approximately 200 examples. Include all retained actions and slot patterns where feasible; record panel IDs. Use official development data only.
- Run CPU inference with Q4 and Q8, measure startup, request latency, invalid outputs, and peak memory. Repeat a short sustained workload.
- Perform one small LoRA smoke run for each candidate, merge, export, and verify that the exported artifact executes on the phone. This is a pipeline check, not final evidence of adaptation benefit.
- Select from feasible candidates using validation joint semantic exact match and the predeclared latency and memory requirements. If neither dominates, prefer the one satisfying requirements with less implementation risk; record the reason.

Freeze the model revision, tokenizer, chat template, reasoning policy, runtime commit, and quantization tools. Cross-model pilot results are descriptive because size, architecture, and training differ.

Backend feasibility gate

CPU is the completion path. Limit GPU investigation to one working day. llama.cpp documents limitations with older A6xx phone drivers [4]. A successful probe must load the chosen quantization, execute representative inputs, produce credible predictions, and log actual offload. If these checks fail, close the GPU branch and preserve the failure report.

Early compute budget

Secure training GPU access in week one and record the spend cap: [currency and amount]. Estimate training and export duration from smoke runs. Estimate inference duration as measured seconds per request multiplied by all planned requests and repeats, with a 30 percent contingency. If the estimate does not fit, simplify optional experiments before final training.

4 Dataset and semantic output protocol

Application rationale and boundary

An offline assistant interprets personal calendar, alarm, and list requests locally. Connectivity independence, local data handling, and integration with device functions motivate edge deployment. Start with typed English requests. The minimum prototype displays a validated action proposal; speech recognition, autonomous execution, cloud accounts, and multi-turn dialogue are outside scope.

MASSIVE subset audit and freeze

Use the en-US locale of a pinned MASSIVE release. The dataset supplies intent and slot annotations with official train, development, and test partitions [5]. Filter those partitions by a versioned productivity allowlist; do not create a new random split or assume subset sample counts before the audit.

- Inventory calendar, alarm, and list actions and associated slot types. Select the smallest coherent set that supports useful local proposals. Report counts by split, action, slot type, and slot combination before freezing coverage.
- Inspect at least 100 development/training examples for ambiguity, annotation quality, missing fields, and action feasibility. Audit duplicates and near duplicates without using test errors for development.
- If coverage is too thin for precise evaluation, broaden only within productivity before G2. If the dataset cannot represent the application, evaluate a bounded PRESTO subset [11] and freeze one primary dataset rather than combining projects.

Canonical semantic representation

Derive one deterministic JSON target from the official intent and annotated slot spans: an intent string and a list of slot objects containing type and verbatim value. Preserve duplicate slots, define list ordering, and represent no slots as an empty list. Version the converter and test round trips. Missing fields must stay missing; never generate fabricated gold arguments.

Convert this benchmark representation into application proposals through a separate mapping. Resolve relative dates with a deterministic component and fixed reference date/timezone. MASSIVE slot extraction does not itself evaluate timestamp normalization, references to unseen personal records, or real action execution.

Baselines and fair output constraints

Compare the untuned SLM and adapted SLM using identical prompts, schemas, decoding, context limits, and constrained JSON decoding where supported. Add a simple rule-based intent/slot parser and a compact encoder jointly trained for intent classification and slot tagging. Intent-only logistic regression may be a diagnostic, but cannot serve as the full-task baseline.

Measure baseline semantic correctness and device cost where deployable. If encoder deployment exceeds its fixed budget, report its quality comparison and label workstation timings as non-comparable. Audit licenses. Keep all final test examples, including empty-slot and malformed-output cases.

Language extension

English is mandatory. PT-BR requires independently validated Brazilian Portuguese data, retained split IDs, and a separate evaluation. MASSIVE pt-PT is European Portuguese. Never mix translated versions of the same source example across training and test partitions.

5 Adaptation and artifact pipeline

Training recipe scaffold

Use supervised fine-tuning with LoRA as the default. QLoRA is a fallback if training memory requires it. Continued pretraining introduces a different objective and corpus and is outside this eight-week minimum.

Setting	Initial choice or decision rule
Objective	Generate canonical intent and slot JSON; mask prompt tokens and optimize target response tokens only.
Adapter rank	Pilot ranks 8 and 16; choose using validation results.
Learning rate	Start at 1e-4; alter only if smoke-run diagnostics justify a recorded change.
Epoch budget	Up to 3; select checkpoint using validation joint semantic exact match.
Effective batch size	Start at 16 using gradient accumulation; adjust for training memory.
Sequence length	Set from tokenizer length audit and prompt; record truncation.
Target modules	Use architecture-appropriate supported linear modules; record exact names and trainable parameter count.
Final seeds	Three fixed seeds, for example 17, 42, and 73; retain all outcomes.

These settings are starting configurations, not validated optima. Limit the development search to two substantive trial runs plus the three final runs; pipeline smoke tests use small subsets. Reserve contingency for failed runs, not an expanding search.

Export and verification sequence

- Train on the workstation GPU and save adapter, optimizer-independent configuration, logs, seed, and selected checkpoint criterion.
- Merge the adapter into the appropriate high-precision base checkpoint. For QLoRA, reload the original base at suitable precision before merging; do not mistake training quantization for deployment quantization [6].
- Evaluate the merged high-precision checkpoint, export to GGUF, and create Q8_0 and Q4_K_M from that same checkpoint.
- Compare outputs on a fixed canary set before and after conversion at matched precision when supported. Investigate major divergence, tokenizer errors, broken templates, and invalid outputs.
- Hash and freeze every released artifact. Export the untuned primary model through the same toolchain.

Artifact count and provenance

The core edge study has eight artifacts: two untuned quantizations plus two quantizations for each of three adapted seeds. High-precision references stay on the workstation. Preserve conversion commands, dependency locks, licenses, model revisions, artifact size, precision scheme, and checksum for each artifact.

6 Android measurement protocol

Device and execution controls

Record physical RAM, exact model identifier, Android and firmware builds, ABI, runtime build flags, backend, offloaded layers, CPU thread count, context size, batch settings, KV-cache type, and model hash. Report available memory separately from installed RAM.

- Use Android NDK cross-compilation and ADB for the initial benchmark. A polished app is optional; a minimal runner that loads a model, accepts requests, and saves logs is sufficient.
- Fix power-saving mode, brightness, screen state, charging state, case usage, and background workload. Record battery level and ambient temperature. Prefer comparable battery ranges and avoid charging during timed sessions when practical.
- Use batch size one and a predeclared request schedule. Keep prompts, decoding, maximum output length, and cache behavior identical across matched configurations.
- Distinguish process-cold startup, first inference, and warmed requests. Do not call a run storage-cold unless OS file-cache state was actually controlled. Record GPU kernel cache state if relevant.

Metrics and instrumentation

Measure	Operational definition
Request latency	Monotonic elapsed time from runner request acceptance through tokenization and output completion; report median and p95.
Startup	Model initialization and first-request duration, separately from steady-state latency.
Memory	Sample process RSS and, where available, Android PSS; retain metric names and sampling interval. Track system available memory.
Reliability	Per-request timeout, crash, load failure, allocation failure, and invalid-output counts.
Thermal behavior	Thermal status and headroom when exposed; sensor temperature only with sensor identity and units. Battery temperature is not CPU temperature [7].
Sustained performance	Latency and request throughput in time windows during a 20-minute fixed workload; compare beginning and end.

Repetitions and workload size

Run quality evaluation on the complete frozen English productivity test subset for each core artifact. Record its audited sample count rather than assuming the full MASSIVE count. For performance, use a fixed approximately 200-request panel across retained actions, slot patterns, and length strata, repeated in three independent sessions per artifact. Randomize artifact order within sessions. Establish a cooldown rule using available thermal readings or a documented fixed rest procedure, then apply it consistently.

Run 20-minute sustained tests for untuned and adapted Q4/Q8 with one preselected adapted seed, in three sessions. Select that seed before final testing. If runtime estimates exceed the available calendar, document any reduced workload before collection; do not silently discard slow or failed configurations.

7 Weekly execution from one through four

Reserve a 30-minute supervisor review each week. Send one page containing completed evidence, unresolved decisions, schedule risk, and the next gate. Update the thesis and experiment log throughout implementation.

Week one October 20 to 26

Budget: 20 hours. Research and requirements 4; device and model pilot 12; compute planning and writing 4.

- Confirm English offline productivity scope, research questions, novelty gap, degree milestones, and training budget. Audit MASSIVE action coverage before selecting pilot examples. Read 8 to 12 closely related papers and build a comparison matrix.
- Inventory the phone and run both candidate models. Perform training/export smoke tests, measure initial CPU performance, and time-box GPU probing.
- Deliver feasibility report, chosen primary model and backend, pilot logs, preliminary execution-time estimate, and thesis outline.

Gate G1: one verified training-to-phone path and a viable compute budget. If neither candidate works, choose a simpler supported SLM within the original size range and repeat the smallest pilot immediately. Do not start full training on an unverified path.

Week two October 27 to November 2

Budget: 20 hours. Data and evaluation 8; baselines 8; protocol and writing 4.

- Freeze official subset manifests, intent/slot schema, annotation converter, prompt, scoring, and decoding. Complete data leakage checks and full-task baseline validation experiments.
- Freeze application requirements, primary comparisons, confidence-interval procedure, timeouts, memory sampling, and session controls.
- Deliver protocol version 1, dataset audit, baseline validation table, and executable evaluation commands.

Gate G2: the evaluation runner scores malformed outputs correctly and produces per-example logs. Final test labels remain outside development decisions. The subset counts and deterministic gold conversion must be complete now.

Week three November 3 to 9

Budget: 20 hours. Training pipeline and trials 12; error analysis 4; writing 4.

- Run the two development trials. Check response-only loss, gradients, trainable parameters, validation learning curves, and output validity.
- Fix the recipe and record why it was selected. Implement encoder deployment within a fixed time budget.

Gate G3: reproducible learning pipeline and fixed final recipe. A weak gain triggers debugging of data and objective; it does not justify test-set tuning.

Week four November 10 to 16

Budget: 20 hours. Final training and exports 12; verification 4; methods writing 4.

- Train three seeds, export all core artifacts, run canary checks, and record hashes. Finish the methods draft and benchmark readiness review.

Gate G4: artifact freeze. Any later repair creates a new version and requires rerunning affected comparisons.

8 Weekly execution from five through eight

Week five November 17 to 23

Budget: 20 hours. Collection supervision 10; validation and preliminary analysis 6; results writing 4. Unattended inference may require substantial additional elapsed time.

- Execute held-out quality evaluation and the three performance sessions. Randomize configuration order and validate logs after each session.
- Save outputs and failures before aggregation. Check sample counts, artifact hashes, timer units, memory units, and missing observations.

Gate G5: complete core quality, latency, and memory evidence. If behind schedule, cancel orchestration and other extensions. Keep slow configurations and failures visible.

Week six November 24 to 30

Budget: 20 hours. Sustained workloads 8; optional policy or robustness 6; analysis and writing 6.

- Complete repeated sustained tests. Analyze latency drift alongside thermal status, power state, and background activity.
- Only after G5, implement the optional memory-based policy and compare it with fixed Q4 and fixed Q8. Record startup and switching costs, sampling overhead, fallbacks, and failures.

Gate G6: evidence collection closes. Explain missing measurements and device limitations. Do not expand to a second domain or device.

Week seven December 1 to 7

Budget: 20 hours. Statistics and figures 8; manuscript and thesis chapters 10; supervisor review 2.

- Compute paired quality differences, confidence intervals, seed variation, session variation, and constraint satisfaction.
- Write the complete manuscript, results chapter, limitations, and conclusion. Choose the strongest supported contribution and remove unsupported claims.
- Produce four core figures: quality versus latency, quality versus peak memory, adaptation gain by precision, and sustained latency over time.

Gate G7: complete draft with traceable claims and a supervisor-reviewed contribution statement.

Week eight December 8 to 14

Budget: 20 hours. Reproduction and fixes 7; revisions 9; submission packaging 4.

- Reproduce selected training/evaluation steps from a clean environment and rerun a representative device benchmark. Check that published aggregates can be regenerated from raw logs.
- Finalize artifact documentation, data/model cards as applicable, licenses, thesis chapter draft, manuscript, and presentation outline. Document an airplane-mode demonstration; report it separately from benchmark semantic accuracy.
- Verify the actual venue call, scope, deadline, page limit, anonymity rules, and publication policy. Obtain the supervisor and coauthor sign-off required for submission.

Gate G8: submission-ready research package. Submit when an appropriate call is open; acceptance and degree defense are separate milestones.

9 Analysis and interpretation

Predeclare the primary analyses

Use joint semantic exact match as the primary metric: the predicted intent and complete slot multiset must match the gold after frozen normalization. Compare parsed semantics, not raw JSON key order. Invalid or incomplete JSON scores zero. Report intent accuracy, exact type/value slot micro-F1, schema validity, and extra or invented slot rates separately. Compute paired adapted-minus-untuned differences for each seed and precision; report all seeds, their mean and spread, and changes in percentage points.

Use paired bootstrap resampling of test examples, stratified by retained action, with the same sampled indices for compared predictions. A practical default is 10,000 bootstrap replicates and a 95 percent interval. For a mean-over-seeds estimate, preserve the paired example sampling across seeds and state that the interval is conditional on the three trained models. Report seed variation separately; three seeds cannot characterize all training randomness.

For device metrics, report per-session summaries and pooled distributions with session boundaries preserved. Do not treat repeated requests as independent physical devices. If estimating latency uncertainty, resample sessions or workload blocks and acknowledge the small number of independent sessions. Report failures as outcomes, not silently omitted timings.

Questions the results must answer

- Is the quality gain positive and large enough to matter under the predeclared application criterion? An interval crossing zero is inconclusive, not proof of equivalence.
- How much quality is lost from high precision to Q8 and Q4? Does the adapted model retain an advantage at each deployed precision?
- Which configurations satisfy quality, latency, memory, and reliability requirements simultaneously? Which lie on the measured Pareto frontier?
- Does the specialist baseline achieve the same task with lower cost? If so, explain where the SLM approach is or is not justified.
- Does sustained performance differ from short-run performance? Describe thermal association without claiming thermal causation unless competing factors are controlled.

Results table scaffold

One row per artifact and session aggregate: model revision | adaptation seed | quantization | backend | joint semantic exact match | intent accuracy | slot F1 | schema validity | invented slot rate | median latency | p95 latency | peak RSS/PSS | startup | failures | sustained change. Use explicit not-measured or unsupported entries rather than invented zeros.

Error analysis and threats to validity

Review a fixed sample of adaptation wins, regressions, invalid outputs, and commonly confused actions and slot values. Discuss benchmark contamination, train-test similarity, prompt sensitivity, limited seed count, quantization implementation, firmware and driver dependence, thermal sensor limitations, and the single-device scope. Add optional diagnostics only if they answer a defined uncertainty; do not search for favorable subsets after seeing test results.

10 Minimal SDK and reproducibility scaffold

Minimum runnable system

Build a thin local runner with capability inspection, model loading, inference, unloading, semantic validation, and telemetry. Keep training, deployment, and evaluation separate. The demo accepts typed English, validates JSON against the action schema, and displays a proposal in airplane mode. Use local test records; real execution is optional.

- inspect_device: physical and available RAM, CPU information, runtime backend enumeration, and available thermal telemetry.
- validate_configuration: probe a model/backend combination and retain supported, unsupported, or unverified status with a reason.
- load and infer: enforce configuration, timeout, output policy, and logging. Record requested backend and actual offload separately.
- select_configuration, optional: use offline profiles plus available memory and a safety margin. Prefer a configuration meeting requirements; fall back or fail clearly if none fits.

Available RAM and model file size are insufficient on their own. Profile model buffers, KV cache, transient allocations, and loading peaks. Avoid rapid switching with a cooldown or hysteresis. For the policy experiment, use a bounded controlled workload and log its memory pressure; compare every policy under the same scenario.

Repository scaffold

Directory	Contents
configs/	Frozen experiment, model, training, and device configurations.
data/manifests/	Split IDs, intent and slot schema, revisions, hashes, and leakage audit.
training/ and export/	Training entry points, adapter merge, conversion, canary checks.
android/ and sdk/	Native runner, device probes, optional configuration policy.
evaluation/ and analysis/	Scoring, benchmark driver, statistics, figures, table generation.
results/raw/ and results/derived/	Immutable request logs and reproducible aggregates.
docs/ and paper/	Protocol, decisions, experiment diary, thesis chapters, manuscript.

Required log fields

run_id, timestamp, code revision, dataset/example ID, model and tokenizer hashes, seed, quantization, backend, device build, context, thread/offload settings, prompt/output token counts, raw and parsed semantic prediction, intent/slot scores, validity, timing fields, memory metric and value, thermal observations, battery/charging state, cache state, timeout/error, and session ID. Never log a measurement as zero when unavailable.

Verification that matters

Test correct parses, wrong intents, missing/extra/repeated slots, empty slots, invalid JSON, and truncation. Verify mappings, converter round trips, split disjointness, actual backend, and units. Regenerate tables from raw logs. Document reproduction commands, hardware, expected outputs, duration, and limitations.

11 Thesis writing and publication

Write the thesis alongside implementation

Chapter	Questions and material to complete
1 Introduction	Problem, application requirements, research questions, scope, and contributions. Draft in weeks 1 and 2; revise after results.
2 Related work	SLM adaptation, quantization, Android inference, semantic parsing metrics, mobile assistant datasets, and specialist alternatives. Explain the specific gap.
3 Methodology	Official filtered splits, schema conversion, baseline fairness, model selection, training recipe, artifact pipeline, metrics, uncertainty, and controls.
4 Implementation	Runner/SDK interfaces, device constraints, measured profiles, deployment path, and optional policy.
5 Results and discussion	RQ-by-RQ tables and figures, error analysis, trade-offs, negative results, and threats to validity.
6 Conclusion	Evidence-supported answers, practical implications, limitations, and feasible follow-up work.

Contribution selection

The preferred contribution is a controlled empirical study of adaptation and deployment precision on an older Android phone, supported by reproducible artifacts. Existing work already examines edge SLMs and domain compression [8, 9]. In week one, record what those studies measure and what this study adds. A different handset alone may provide insufficient novelty. The adaptation interaction, semantic reliability under quantization, sustained workload evidence, or actionable findings must carry the contribution. Describe this as productivity task adaptation; do not claim acquisition of personal calendar knowledge from fine-tuning.

- Claim A, if supported: adaptation produces a practically meaningful gain that survives deployment quantization.
- Claim B, if supported: particular configurations meet requirements while others fail, producing useful deployment guidance.
- Claim C, if supported: specialist baselines dominate or adaptation gains disappear; explain this result without overstating generality.

Publication sequence

Prepare one coherent manuscript and a reusable artifact release. Start with an efficiency, edge AI, or evaluation workshop, or an eligible student research workshop. Consider an application or industry track when the deployment evidence fits its call. Do not divide the same small result into overlapping submissions.

In week seven, inspect current official calls and choose a primary and backup venue. Record deadline, scope, page limit, anonymity, artifact policy, and preprint policy. EACL 2027 Industry Track closed submissions on September 11, 2026, so it is not an available destination for this December package [10]. No future acceptance or open workshop call is assumed.

Paper outline

Abstract; introduction and contributions; related work; experimental method; implementation and device protocol; results; discussion and limitations; conclusion. Each claimed contribution must point to a comparison, table, or released artifact. Finalize the abstract only after the results are fixed.

12 Decision gates and completion checklist

Risk or trigger	Required response
No verified export by G1	Simplify model/runtime immediately. Repeat the smallest end-to-end pilot before training.
GPU fails its one-day probe	Proceed CPU-only; preserve compatibility evidence and close the branch.
Training exceeds budget	Reduce validation search and optional runs; retain matched core comparisons and disclose deviations.
No adaptation improvement	Audit labels, response masking, prompts, and learning curves using validation data. If correct, retain the negative result.
Week five evidence incomplete	Cancel policy, extra models, energy, and extra diagnostics. Finish the primary matrix.
Device unstable or memory-limited	Use the predeclared timeout/failure protocol; adjust only through a documented protocol version and rerun affected comparisons.
Supervisor requires broader evidence	Negotiate the thesis scope or calendar explicitly. Do not quietly add another domain or device to the fixed schedule.

Final research completion checklist

- Research questions and minimum practical effect are documented; model/domain selection and protocol deviations are traceable.
- Data audit, final training seeds, model hashes, runtime commit, and exact phone firmware are recorded.
- Core quality, performance, sustained-load, and specialist baseline results are available with uncertainty and failure counts.
- Every table and figure regenerates from saved logs. A clean-environment reproduction check is documented.
- Conclusions distinguish measured findings from hypotheses and explain single-device and task limitations.
- Thesis chapter draft and manuscript are reviewed; artifact licensing and redistribution permissions are checked.
- University format, submission forms, examination timeline, and supervisor requirements are tracked separately.

Weekly review template

Week and date: [ ]. Evidence completed: [artifact IDs and links]. Current gate: [pass / at risk / fail]. Findings: [measured facts]. Blocker and next action: [ ]. Hours spent and remaining: [ ]. Protocol changes and rationale: [ ]. Supervisor decisions: [ ]. Next deliverable and due date: [ ].

First 48 hours

Record Android and firmware details; obtain training GPU access; create the repository and experiment diary; download permitted candidate checkpoints; execute one CPU Q4 inference on the phone; schedule the first supervisor scope review. Use that first measured run to refine the benchmark time budget.

13 Sources and evidence register

Sources checked during planning on October 20, 2026. Runtime and model documentation changes; pin exact revisions when implementing. The procedures, time budgets, and starting hyperparameters in this plan are research design recommendations, not claims of measured performance.

[1] Realme 7 Pro specifications. Official device chipset and GPU specification.

https://www.realme.com/mea-en/realme-7-pro/specs

[2] Qwen3 0.6B model card. Primary older candidate; pin weights, tokenizer, and template.

https://huggingface.co/Qwen/Qwen3-0.6B

[3] Qwen3.5 0.8B model card. Newer candidate, hybrid architecture, and task-specific fine-tuning use.

https://huggingface.co/Qwen/Qwen3.5-0.8B

[4] llama.cpp OpenCL backend. GPU support and older Adreno driver limitations.

https://github.com/ggml-org/llama.cpp/blob/master/docs/backend/OPENCL.md

[5] MASSIVE official dataset repository. Intent and slot annotations, source IDs, locales, and official partitions.

https://github.com/alexa/massive

[6] Hugging Face PEFT LoRA documentation. Adapter merging and standalone model deployment.

https://huggingface.co/docs/peft/main/package_reference/lora

[7] Android Thermal API. Thermal status, headroom, and device limitations.

https://developer.android.com/games/optimize/adpf/thermal

[8] Demystifying Small Language Models for Edge Deployment. ACL 2025 related work and positioning.

https://aclanthology.org/2025.acl-long.718/

[9] Compressing Language Models for Specialized Domains. EACL 2026 related work on specialized-domain compression.

https://aclanthology.org/2026.eacl-long.347/

[10] EACL 2027 Industry Track call. Deployment-study expectations and closed submission deadline.

https://2027.eacl.org/calls/industry/

[11] PRESTO official dataset repository. Contingency dataset for contextual assistant parsing, subject to a bounded feasibility audit.

https://github.com/google-research-datasets/presto

Literature matrix to complete in week one

For each related paper, record citation, task and language, model and size, adaptation method, quantization, physical device, backend, quality measures, systems metrics, repetitions, released artifacts, limitations, and the precise difference from this study. Resolve the novelty statement with the supervisor before full experimental investment.